CBOT SOYBEAN OIL FUTURES (ZL)

Spread & Curve Structure — Fundamental Reference

From Cooking Oil to Renewable Diesel Feedstock — the Crush, Biofuel Policy, and the Global Vegetable Oil Complex

This document presents a fact-based reference on the fundamental drivers of CBOT Soybean Oil futures (ticker: ZL), contract size 60,000 lb, one of the two co-products (alongside soybean meal, ZM) generated by crushing soybeans, and the contract most directly exposed to one of the most significant demand-side structural shifts in the agricultural complex over the past several years: the rapid growth of renewable diesel and biodiesel demand for vegetable oils as transportation biofuel feedstocks. Where soybean meal's fundamentals are tied predominantly to the livestock and poultry feed cycle described in the CBOT Soybeans reference document, soybean oil sits at the intersection of three distinct demand worlds: traditional food use (cooking oil, shortening, and food manufacturing), the rapidly growing biofuel sector (connecting soybean oil to crude oil, diesel, and federal and state biofuel policy), and the broader global vegetable oil complex, in which soybean oil competes for substitutable demand with palm oil, canola/rapeseed oil, and other vegetable oils produced in entirely different geographies. This document presents the soybean oil-specific fundamentals in full, with particular emphasis on the crush relationship to soybeans and meal, the biofuel policy framework now driving a structural share of demand, and the cross-market relationships to crude oil, palm oil, and the broader vegetable oil complex that distinguish soybean oil from the other agricultural contracts covered elsewhere in this series.

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1. Contract Specifications and the Crush Relationship

CBOT Soybean Oil is one of the two co-products of the soybean crush (alongside Soybean Meal, ZM), priced and traded as its own dedicated futures contract rather than as a derivative calculation from the soybean (ZS) price alone.

Parameter	Detail
Exchange	CME Group — Chicago Board of Trade (CBOT)
Ticker	ZL (front month)
Underlying	Crude (unrefined) soybean oil, in tank car or bulk loading
Contract size	60,000 lb (approximately 27.2 metric tonnes)
Quotation	US cents per pound (¢/lb); minimum tick = 1/100 cent/lb = $6.00 per contract
Contract months	January (F), March (H), May (K), July (N), August (Q), September (U), October (V), December (Z) — eight delivery months per year, the densest calendar of the three soybean-complex contracts (ZS has seven; ZM shares this same eight-month structure), reflecting both high liquidity and processors' need for granular forward hedging coverage across the calendar year.
Settlement type	Physical delivery via shipping certificate at exchange-approved tankage, predominantly in the Illinois/Chicago-area processing and storage network — a liquid-bulk handling and storage system structurally different from the dry-grain elevator delivery system described for the grain contracts elsewhere in this series.
Delivery grade	Crude (unrefined) soybean oil meeting exchange-specified quality parameters (free fatty acid content, moisture and insoluble impurities, colour) — this is the oil as it emerges directly from the crushing/extraction process, prior to the food-industry refining, bleaching, and deodorising (RBD) steps or biofuel-specific pretreatment described in Section 4.
Trading hours	CME Globex: Sunday–Friday 19:00–07:45 ET and 08:30–13:20 ET, the same general electronic-session structure as the other soybean-complex contracts.
Last trading day	Business day prior to the 15th calendar day of the delivery month
Contract value at $0.45/lb	$27,000 per contract
Related contracts	Soybeans (ZS) and Soybean Meal (ZM) — together with ZL, these constitute "the soybean complex" and the three legs of the board crush described in Section 3 and in the CBOT Soybeans reference document.

1.1 Crude Oil, Not Refined Oil — Why the Distinction Matters

- The ZL contract prices crude (unrefined) soybean oil, the immediate output of the crushing/extraction process at a processing plant, rather than the refined, bleached, and deodorised (RBD) oil that ultimately reaches retail food shelves or food manufacturing customers, or the further-processed/pretreated oil used as biofuel feedstock. The refining margin (the cost and value added in converting crude oil to RBD oil) is a separate, downstream economic layer not directly priced by ZL itself, analogous in concept (though a distinct specific mechanism) to the raw-versus-refined distinction described for sugar (No. 11 versus White Sugar No. 5) elsewhere in this document series.

2. Why Soybean Oil Is a Co-Product, Not a Standalone Crop

Unlike every other commodity covered elsewhere in this document series, soybean oil has no independent agricultural production process of its own: it does not grow, is not planted, and is not harvested. Its entire supply is generated as one of two outputs of crushing soybeans, meaning soybean oil's supply-side fundamentals are inherited directly from the soybean crop fundamentals described in full in the CBOT Soybeans reference document, filtered through the crushing industry's processing capacity and output-mix decisions.

2.1 The Fixed-Ratio Yield From Crushing

- Crushing a 60 lb bushel of soybeans yields approximately 11 lb of crude soybean oil and 44–48 lb of soybean meal (plus hulls and processing losses), a ratio that varies only modestly with bean quality, oil content genetics, and processing technology. Because this yield ratio is largely fixed by the biology of the soybean itself, soybean oil supply cannot be substantially increased independent of total soybean crush volume — a processor cannot simply choose to produce materially more oil relative to meal from a given quantity of beans without a change in soybean variety/genetics, which operates on an agricultural breeding timescale, not a short-term production-planning timescale.
- **The practical implication**: soybean oil supply growth is, over any near-to-medium-term horizon, a function of total soybean crush volume growth (Section 3) rather than of oil-specific supply decisions, meaning a trader assessing the soybean oil supply outlook must look at the same soybean crop and crush-capacity fundamentals described in the CBOT Soybeans reference document, not at any oil-specific agricultural variable.

2.2 The Co-Product Revenue Allocation

- Because meal and oil are joint products of a single crushing process, neither product's "production cost" can be cleanly separated from the other's; rather, the combined revenue from selling both co-products, relative to the cost of the soybean input, determines overall processing profitability (the crush margin, detailed in Section 3). This means soybean oil's price is not set by an independent oil-specific cost-of-production calculation in the way, for example, that a mined commodity's price relates to extraction cost — it is one of two outputs whose combined value must clear the crush margin required to sustain processing activity.

3. The Crush — Revisited From the Oil Side

The crush relationship between soybeans, meal, and oil is described in detail in the CBOT Soybeans reference document; this section summarises that relationship from the specific perspective of the oil leg and its growing relative importance to overall crush economics.

3.1 The Board Crush

- The crush margin — combined meal and oil revenue per bushel crushed, less the soybean input cost — can be calculated directly from the three CBOT contract prices (ZS, ZM, ZL) using the fixed yield ratio described in Section 2.1, a calculation referred to in the trade as the "board crush." Processors hedge their actual physical crush margin using offsetting ZS/ZM/ZL futures positions, and the board crush is also directly traded by speculative participants taking a view on processing economics independent of any single leg's flat-price direction.

3.2 The Oil Share of Crush Value — A Structural Shift

- Historically, soybean meal represented the larger share of total crush revenue, reflecting meal's larger output weight per bushel (Section 2.1) and its role as the primary protein source for the large livestock and poultry feed market. Over recent years, the rapid growth in biofuel-driven soybean oil demand (Section 4) has, at various points, increased oil's relative contribution to total crush value compared to its historical share — a structural shift tracked closely by processors and analysts, since it affects not only the crush margin's overall level but the relative sensitivity of total crush profitability to oil-specific versus meal-specific demand developments.
- **Implications for processing investment**: as the oil share of crush value has grown, US soybean processing capacity investment decisions have, per industry reporting, increasingly been justified by oil-demand (and specifically biofuel-demand) growth expectations, a notable shift from a processing industry historically built out primarily to serve meal/feed demand growth.

4. Biofuel Demand — The Structural Growth Story

The rapid growth of renewable diesel and biodiesel demand for soybean oil is the single most important fundamental development in the soybean oil market over the past several years, and the primary reason soybean oil requires its own dedicated analytical framework distinct from a simple "co-product of the soybean crush" treatment.

4.1 Biodiesel Versus Renewable Diesel — A Necessary Distinction

- **Biodiesel**: produced via a transesterification process (reacting vegetable oil or animal fat with an alcohol, typically methanol) to produce fatty acid methyl esters (FAME). Biodiesel is chemically distinct from petroleum diesel and is typically blended with it at varying percentages (commonly labelled, e.g., B5, B20, B100) rather than serving as a complete drop-in replacement.
- **Renewable diesel**: produced via a hydrotreatment process (chemically similar in concept to petroleum refining) that converts vegetable oil or animal fat into a fuel that is chemically nearly identical to petroleum diesel — a genuine "drop-in" substitute usable in existing diesel engines and infrastructure without blending limitations, in contrast to biodiesel's blend-percentage constraints.
- **Why the distinction matters for soybean oil demand**: because renewable diesel is a more flexible, less blend-constrained substitute than biodiesel, the rapid build-out of US renewable diesel refining capacity over recent years (including both new-build facilities and the conversion of some existing petroleum refining capacity to renewable diesel production) has been a particularly significant structural demand growth driver, since renewable diesel capacity additions translate more directly into incremental vegetable oil feedstock demand than biodiesel's blend-percentage-constrained growth path.

4.2 The Policy Framework

- **The Renewable Fuel Standard (RFS) — Biomass-Based Diesel category**: as referenced in the CBOT Soybeans reference document and described in fuller detail (from the ethanol/RIN side) in the RBOB reference document, the RFS sets annual Renewable Volume Obligations across several fuel categories, including a biomass-based diesel category covering both biodiesel and renewable diesel. EPA's annual RVO-setting process for this category is a direct policy lever on aggregate biofuel-feedstock demand, including soybean oil demand specifically.
- **Blender's Tax Credit and the 45Z Clean Fuel Production Credit**: federal tax credit policy supporting biodiesel and renewable diesel production and blending has, at various points, included a per-gallon blender's tax credit and, more recently, a production-based clean fuel credit (45Z, introduced under the Inflation Reduction Act framework) that ties credit value to a fuel's calculated carbon intensity — a structural change from a flat per-gallon credit to a carbon-intensity-differentiated credit, which can affect the relative economics of different feedstocks (including soybean oil relative to used cooking oil, animal fats, and other lower-carbon-intensity feedstocks) depending on each feedstock's calculated carbon intensity score.
- **State-level Low Carbon Fuel Standard (LCFS) programmes**: California's LCFS (and similar programmes adopted by other US states) creates a tradeable credit market rewarding lower-carbon-intensity transportation fuels, providing an additional (state-level, layered on top of federal policy) economic incentive for renewable diesel and biodiesel production and, by extension, for vegetable oil feedstock demand in states participating in such programmes.

4.3 Feedstock Competition — Soybean Oil Versus Other Feedstocks

- Soybean oil competes with other biofuel feedstocks — used cooking oil (UCO), animal fats and tallow, canola oil, and (particularly relevant given import competition dynamics) imported feedstocks including used cooking oil and other waste-oil-derived feedstocks from various countries — for use in US biodiesel and renewable diesel production. Because different feedstocks carry different calculated carbon intensity scores under programmes like the federal 45Z credit and California's LCFS (waste-oil-derived feedstocks generally score more favourably than virgin vegetable oils including soybean oil, since they avoid the land-use and cultivation emissions associated with growing a dedicated oil crop), feedstock policy treatment is a relevant variable affecting soybean oil's relative competitiveness as a biofuel feedstock, independent of its flat-price level alone.

5. Traditional Food Demand

Despite the rapid growth of biofuel demand described in Section 4, traditional food use remains a substantial component of soybean oil demand, both domestically and globally, and provides a more stable, less policy-dependent demand base than the biofuel sector.

5.1 Domestic Food Uses

- Soybean oil is used in cooking oil (retail and food-service), shortening, margarine, salad dressings, and as a frying and baking ingredient across the food manufacturing sector. US retail and food-service demand for soybean oil specifically (as opposed to other vegetable oils) reflects both its relative price and its functional and nutritional characteristics (including the trans-fat-related reformulation of partially hydrogenated soybean oil products that occurred industry-wide following regulatory action against partially hydrogenated oils in prior years, a structural demand-composition shift now largely complete rather than an ongoing variable).

5.2 Export Demand for Food Use

- US soybean oil is exported for food use to a range of destinations, though export volumes for crude soybean oil specifically (as opposed to the much larger raw soybean export trade described in the CBOT Soybeans reference document) are comparatively modest, reflecting both strong domestic US demand (food plus, increasingly, biofuel) absorbing a large share of US oil production, and the broader pattern in global vegetable oil trade in which oil is more commonly traded in refined form closer to the point of final consumption, or sourced from origins (notably palm oil from Indonesia/Malaysia) with lower production costs for pure food-oil applications.

6. The Global Vegetable Oil Complex — Cross-Oil Substitution

Soybean oil is one of several major vegetable oils that compete, to varying degrees, for substitutable food and biofuel-feedstock demand, making the broader vegetable oil complex — and palm oil specifically — a central cross-market consideration for soybean oil.

Oil	Approx. Share of World Vegetable Oil Production	Primary Geography	Key Characteristics	Relevance to ZL
Palm oil	~35–40%, the largest single vegetable oil by volume	Indonesia and Malaysia (together the dominant majority of world production)	Highest yield per hectare among major oil crops; semi-solid at room temperature; widely used in food manufacturing (confectionery, packaged foods) and, in Indonesia and Malaysia specifically, mandated domestic biodiesel blending programmes	The single most important cross-oil substitution relationship for soybean oil; palm oil price moves (driven by Indonesian/Malaysian production, weather, and biodiesel mandate policy) directly affect soybean oil's relative competitiveness in both food and biofuel applications
Soybean oil	~25–28%	United States, Brazil, Argentina (mirroring the soybean production geography described in the CBOT Soybeans reference document)	Liquid at room temperature; the dominant vegetable oil in US food use historically; now also a primary US biofuel feedstock (Section 4)	This document's subject — the second-largest vegetable oil globally by volume
Canola/rapeseed oil	~12–14%	Canada, European Union	Relatively low saturated fat content; a significant biodiesel feedstock in the EU specifically, under EU biofuel policy	A competing feedstock and, in some applications, a substitutable food oil; Canadian and EU canola production and biofuel policy are relevant secondary variables
Sunflower oil	~8–10%	Ukraine, Russia, EU (historically Ukraine and Russia together a dominant share of world exports)	A significant food-oil substitute, particularly in European and some Asian markets	War-related disruption to Ukrainian sunflower oil exports since February 2022 has, at various points, affected the broader vegetable oil complex's supply balance and relative pricing, indirectly relevant to soybean oil
Other (corn oil, cottonseed oil, others)	Remainder	Varies	Smaller, more specialised applications	Collectively a minor but non-zero contributor to total substitutable vegetable oil supply

6.1 The Palm-Soy Oil Spread

- The price spread between palm oil (typically referenced via the Malaysian Bursa Malaysia Derivatives crude palm oil contract, or FCPO) and soybean oil (ZL) is a closely tracked relative-value relationship, reflecting the partial substitutability of the two oils in food manufacturing applications and, increasingly, in biofuel feedstock use where policy treatment allows. A wide palm-soy discount (palm oil cheap relative to soybean oil) generally supports palm oil's share of substitutable food-use demand at the margin, while a narrow or inverted spread can shift demand back toward soybean oil.

6.2 Indonesian and Malaysian Policy as a Cross-Market Variable

- Indonesia and Malaysia, as the dominant world palm oil producers, periodically adjust export tax and (in Indonesia's case, at various points) export licensing or outright export restriction policy on palm oil, generally aimed at managing domestic cooking oil price stability or supporting the countries' own mandated domestic biodiesel blending programmes (Indonesia's B30/B35-type mandates being a notable example). Such policy changes affect global palm oil export availability and price, with knock-on effects on the palm-soy spread and, at the margin, on soybean oil demand and pricing.

7. The Forward Curve Structure

ZL's forward curve reflects the same general cost-of-carry framework as other storable agricultural commodities (storage, financing, and insurance costs for holding crude soybean oil), modulated by the underlying soybean crush cycle (inherited from the CBOT Soybeans calendar) and, increasingly, by the calendar of biofuel policy events described in Section 4.2.

7.1 Carry Inherited From the Crush Cycle

- Because soybean oil supply is generated continuously by ongoing crush activity (rather than by a single discrete annual harvest event the way the underlying soybean crop is), ZL's own curve does not exhibit as sharply defined a harvest-driven seasonal pattern as ZS itself; instead, ZL's curve reflects a blend of the underlying soybean supply seasonality (via crush input availability and cost) and oil-specific demand-side seasonality (described in Section 7.2).

7.2 Biofuel Policy Calendar Effects

- EPA's RVO-setting process and finalisation timeline for the biomass-based diesel category (Section 4.2), and the periodic legislative or regulatory calendar around blender's tax credit or 45Z credit policy renewal/modification, can introduce curve-specific positioning ahead of known or anticipated policy decision dates, a demand-side calendar effect with no direct parallel in the underlying soybean crop's own seasonal pattern.

7.3 The Crush Spread's Influence on the ZL Curve Specifically

- Because processors actively manage forward crush margin hedging (Section 3.1) across multiple delivery months simultaneously, commercial hedging flow in ZL is frequently linked to simultaneous offsetting activity in ZS and ZM at the same forward dates, meaning ZL curve positioning at any given point in time should be assessed alongside the corresponding points on the ZS and ZM curves rather than in isolation.

8. Macro and Cross-Market Drivers

8.1 Crude Oil and Diesel Prices

- As described in Section 4, the biofuel demand pillar connects soybean oil directly to crude oil and diesel pricing: higher diesel prices generally improve the economics of blending biodiesel or renewable diesel (produced from soybean oil and other feedstocks) into the fuel supply, supporting feedstock demand at the margin, though — as with corn ethanol's relationship to the RFS mandate described in the Corn reference document — the RFS biomass-based diesel mandate and state LCFS programmes provide a policy-driven demand floor that partially decouples biofuel feedstock demand from pure market-based fuel-price economics.

8.2 The Soybean Crush Cycle (Inherited From ZS)

- As described in Sections 2–3, soybean oil's supply-side fundamentals are inherited from the underlying soybean crop and crush-capacity-utilisation cycle described in full in the CBOT Soybeans reference document — the US planting/growing/harvest calendar, the Brazilian growing season's offsetting six-month supply pulse, and NOPA monthly crush data are all directly relevant background for ZL even though they are not oil-specific variables.

8.3 Palm Oil and the Broader Vegetable Oil Complex

- As detailed in Section 6, palm oil price moves and Indonesian/Malaysian production and policy developments are a primary cross-market input to soybean oil via the substitution relationship described in Section 6.1.

8.4 US Dollar

- As a globally traded, USD-denominated commodity, soybean oil exhibits the standard dollar-priced-commodity relationship described elsewhere in this document series, though the relative importance of this channel is somewhat reduced for ZL compared to the raw soybean (ZS) contract, given crude soybean oil's comparatively modest direct export volume (Section 5.2) relative to the much larger raw soybean export trade.

8.5 Speculative Positioning — COT Data

- **CFTC COT for CBOT Soybean Oil**: weekly managed money net positioning is tracked, consistent with the positioning framework described throughout this document series, and — as noted in the CBOT Soybeans reference document — is frequently monitored alongside the related ZS and ZM positioning data given the crush-margin relationship connecting all three contracts.

9. Documented Seasonal Patterns

ZL's seasonal pattern is a composite of the underlying soybean supply seasonality (inherited from ZS, including the US-Brazil two-hemisphere calendar) and oil-specific demand-side seasonal and policy-calendar effects.

Period	Key Driver	Mechanism	Relevant Contracts
Jan–Feb	US post-harvest crush activity continuing at typically strong utilisation; early Brazilian crop outlook beginning to factor into forward expectations	Crush margin economics and NOPA monthly data are the primary near-term signal	January, March contracts
Mar–May	Brazilian harvest confirms the Southern Hemisphere contribution to global soybean (and therefore oil) supply	Brazilian crush capacity utilisation and export competition (including from Argentine crush, given Argentina's outsized meal/oil export role described in the CBOT Soybeans reference document) become relevant	May, July contracts
Mid-year	US new-crop growing-season weather (inherited risk from ZS, see Section 8.2); EPA RVO-setting process for the upcoming compliance year often advances during this period	Both agricultural-weather and biofuel-policy-calendar effects can be simultaneously relevant	August, September contracts
Year-end	US harvest-driven crush capacity ramp-up; biofuel policy compliance-year transitions; tax credit/RVO finalisation news around year-end is a recurring policy-calendar catalyst	New-crop crush margin economics; any year-end biofuel policy finalisation or renewal news	October, December, January contracts

Because soybean oil has no agricultural growing season of its own, its seasonal pattern is less calendar-certain in the way corn's pollination window or wheat's heading-stage window are; it is instead a composite signal inherited from the underlying soybean crop calendar plus an increasingly significant policy-calendar layer tied to the biofuel mandate and tax credit framework described in Section 4.

10. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Typical Timing	Primary ZL Curve Impact
WASDE (Soybean Oil component)	USDA WAOB	Monthly	~11th–12th of month, 12:00 ET	All contracts — includes the oil-specific demand breakdown (food use, biofuel use, exports, ending stocks)
NOPA Monthly Crush Report	National Oilseed Processors Association	Monthly	~3rd–5th business day of month	Front contracts — oil production and stocks, a higher-frequency complement to USDA figures
EPA Renewable Fuel Standard RVO proposals/finalisations	US EPA	Annual (proposal then final rule)	Variable, often mid-year for proposal, year-end for final	All contracts — direct policy lever on biomass-based diesel category demand
45Z Clean Fuel Production Credit guidance/updates	US Treasury / IRS	As issued	Variable	All contracts — affects relative feedstock economics (Section 4.3)
California LCFS credit price and programme updates	California Air Resources Board (CARB)	Continuous (credit market) / periodic (programme updates)	Variable	Biofuel-demand-relevant background; affects renewable diesel economics in the LCFS-covered market specifically
Malaysian Palm Oil Board (MPOB) monthly data	MPOB	Monthly	~10th of month	All contracts — Malaysian palm oil production/stocks/export data, the primary cross-oil substitution signal
Indonesian palm oil export policy announcements	Indonesian Ministry of Trade and related agencies	As scheduled/amended	Variable	All contracts — affects global palm oil export availability and the palm-soy spread
CONAB / Bolsa de Cereales (Brazilian/Argentine soybean components)	CONAB; Bolsa de Cereales de Buenos Aires	Monthly/weekly	Variable	Inherited from ZS — Southern Hemisphere soybean supply feeding into crush and oil output
CFTC Commitments of Traders (Soybean Oil)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — tracked alongside ZS and ZM given the crush-margin relationship

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CBOT Soybean Oil occupies a unique position among the contracts covered in this document series: it is the only major agricultural futures contract whose entire supply is generated as a processing co-product rather than through an independent planting-to-harvest agricultural cycle of its own, meaning its supply-side analysis is inherited wholesale from the soybean crop and crush-capacity fundamentals described in the CBOT Soybeans reference document. Its demand side, by contrast, has become increasingly distinct from soybean meal's feed-driven demand profile, owing to the rapid structural growth of renewable diesel and biodiesel demand under the federal RFS biomass-based diesel mandate, the 45Z clean fuel production credit, and state-level programmes including California's LCFS — a policy-driven demand transformation with genuine parallels to corn ethanol's relationship to the RFS, but operating through a different fuel category and feedstock-competition framework.

For spread traders, the most reliable analytical framework treats ZL as two layered exposures: a soybean-crush-derived supply exposure (best tracked via the same ZS/Brazilian-calendar fundamentals described in the CBOT Soybeans reference document, plus NOPA monthly crush data), and an increasingly important biofuel-policy-driven demand exposure (best tracked via the EPA RVO calendar, 45Z and LCFS credit policy, and feedstock competition dynamics). The board crush margin and the palm-soy spread are the two most important cross-market relationships connecting ZL to, respectively, the rest of the soybean complex and the broader global vegetable oil market.